PASUPATHI RAGAVAN T

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Personal Details

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Objective

Electronics and IoT Engineer with hands-on experience in embedded systems, real-time monitoring, hardware-software integration, and automation projects. Skilled in Python, Raspberry Pi, ESP8266, REST APIs, cloud data logging, and sensor-based systems. Built AI-integrated projects involving anomaly detection, LLM APIs, and real-time alert systems using OpenAI and Groq APIs. Strong interest in transitioning into AI engineering, intelligent automation, and backend-driven AI solutions.

Experience

- **Enthu Technology** — 31/07/2023 - 07/08/2023
  Student Trainee
  - Built real-time environmental monitoring prototypes using Raspberry Pi, ESP32, and LoRa modules with cloud data logging — simulating IoT-to-cloud data pipeline architectures.
  - Implemented sensor fusion workflows, collecting, processing, and visualizing structured data streams from multiple hardware sensors simultaneously.
  - Gained foundational experience in embedded programming, communication protocols (UART, SPI, I2C), and event-driven system design.
- **Imai Technologies** — 16/04/2025 - Present
  Embedded Support Engineer
  - Designed and deployed IoT-based automation systems integrating ESP8266, NodeMCU, and Raspberry Pi with cloud platforms for real-time data ingestion and monitoring.
  - Configured Raspberry Pi GPIO interfaces to control relay arrays, sensors, and actuators — enabling rule-based automation logic analogous to AI inference pipelines.
  - Programmed Wi-Fi-enabled microcontrollers for device-to-cloud data transfer, reducing manual monitoring effort by enabling continuous remote telemetry.
  - Performed systematic fault detection and root-cause analysis on electronic systems, applying structured debugging methodologies transferable to AI/ML model validation.
  - Assisted in end-to-end field deployment of smart systems, coordinating hardware setup, firmware flashing, and system testing..

Education

- **Coimbatore Institute of Engineering and Technology** — 2025
  B.E - Electronics and Communication Engineering
  7.35%

Skills

- Embedded & IoT: Raspberry Pi, ESP8266, NodeMCU, Arduino
- Systems & Tools : Sensor interfacing, Data monitoring, cloud Data Logging, Python HTTP Server
- AI/ML Tools : Python, OpenAI API, Groq API, Anomaly Detection, JSON
- Programming: C, C++ , Python
- Communication Protocols: UART, SPI, I2C, Wi-Fi
- Concepts: Real-Time Monitoring, Data Pipelines , Statistical Anomaly Detection(Z - Score/ 2 -Sigma) , LLM API

integration, Automation Systems

Projects

- **IoT Sensor Data Explainer AI Powered Diagnostic Pipeline**
  - Built an end-to-end AI data pipeline in Python that ingests IoT sensor CSV data, applies statistical anomaly detection (z-score/2-sigma algorithm), and sends structured results to an LLM (GPT-4o-mini / LLaMA 3.3 via Groq API) using engineered system prompts to generate plain-English diagnostic reports.
  - Designed a prompt engineering workflow that enforces structured JSON output from the LLM, enabling reliable parsing and downstream rendering in a custom dark-mode web dashboard with real-time charts — demonstrating full-stack AI application development
- **Smart IoT Alert System - Real Time AI Alert Engine**
  - Built an AI-powered real-time alert system that reads live IoT sensor data, classifies severity as CRITICAL / WARNING / NORMAL using Groq LLM API with engineered prompts, and generates plain-English technician alerts with recommended actions — simulating a production IoT monitoring pipeline.
  - Implemented automated alert logging to JSON, multi-level severity detection across 4 sensor channels (temperature, humidity, vibration, voltage), and tested with 3 real-world failure scenarios including motor overheat (72°C) and voltage drop — demonstrating end-to-end AI integration skills.
- **IoT Based Real - Time Bridge Structural Health Monitoring System**
  - Engineered a distributed sensor network using IoT devices, cloud computation, and mobile alerting to deliver predictive structural health data — demonstrating skills in data pipeline design, cloud integration, and automated alert systems relevant to AI monitoring solutions.
- **Smart Boom Barrier Position Detection Using MPU6050 and NodeMCU**
  - Developed an intelligent edge-detection system using accelerometer/gyroscope sensor data and a threshold-based classification algorithm to determine real-time barrier states (open/closed/transitioning) — directly analogous to supervised classification logic in ML models.
- **IoT Based Circuit Breaker Fault Detection & Remote Control System**
  - Built an automated fault detection and system-reset solution for power distribution lines using IoT sensors and a real-time decision engine — demonstrating experience in rule-based AI automation and event-driven system design
- **Automatic Rain Sensing Wiper Control System (Arduino)**
  - Designed a sensor-triggered, state-machine-driven actuator system using an Arduino microcontroller to autonomously respond to environmental inputs — foundational to understanding reinforcement-style feedback loops in AI systems.

Achievements & Awards

- Attended National Conference on CIET
- Attended smart India Hackathon in CIET
- Attended Hands-on workshop in PCB Designing using Easy EDA.

Languages

- Tamil
- English

DECLARATION

- I hereby declare that the above furnished information and details
  are genuine and true to my knowlwdge.